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Technical Appendix: Comparative Propulsion Current-Draw Analysis

2812 1100 KV, 6S, 9×4.0 Three-Blade Benchmark with Theoretical Scaling

Prepared for technical documentation and GitHub repository upload

1. Objective

This appendix develops a current-draw model for a benchmark propulsion system composed of a 2812 1100 KV motor, a 24.0 V supply, and a 9×4.0 three-blade propeller. The section begins with the directly measured electrical data and then links those measurements to classical propeller power laws. The aim is not to claim an exact flight-prediction tool, but to provide a rigorous academic comparison between measured static behaviour, theoretical scaling, a parametric inflight estimate, and a two-blade comparison framework.

2. Directly measured benchmark data

The measured benchmark data for the propulsion unit are as follows:

- Motor: 2812, 1100 KV
- Supply voltage: 24.0 V
- Battery class: 6S
- Propeller: 9×4.0, 3-blade
- Measured electrical input power: 1150 W to 1250 W

From the electrical relation P = VI, the current range is obtained as follows.

$$I = P / V$$

$$I_min = 1150 / 24.0 = 47.92 \text{ A}$$

$$I_max = 1250 / 24.0 = 52.08 \text{ A}$$

Hence the measured static current lies between 47.9 A and 52.1 A. The midpoint current is 50.0 A, corresponding to a midpoint electrical input power of 1200 W.

3. No-load electrical speed limit

The no-load rotational speed upper bound is determined from the motor KV constant and the applied voltage.

$$RPM_nl = K_V \cdot V$$

$$RPM_nl = 1100 \cdot 24.0 = 26400 \text{ rpm}$$

$$n_nl = 26400 / 60 = 440 \text{ rev/s}$$

This is not the actual loaded speed. It is only the electrical no-load upper bound. The real operating speed under propeller load is lower. A loading factor is therefore introduced.

$$\lambda = n / n_{nl}$$

$$n = \lambda n_{nl}$$

In RPM form, the loaded speed can be written as RPM = $\lambda \cdot 26400$.

4. Why current rises approximately with the cube of RPM

The classical shaft-power relation for a propeller is given by the power-coefficient form below.

$$P_{shaft} = C_P \rho n^3 D^5$$

$$P_e = P_{shaft} / \eta$$

$$I = (C_P \rho n^3 D^5) / (\eta V)$$

$$I \propto n^3$$

$$I_2 / I_1 = (n_2 / n_1)^3$$

Thus, even a modest RPM increase causes a much larger current rise. A 5% RPM increase produces a current multiplication of $1.05^3 = 1.158$, or about a 15.8% rise. A 10% RPM increase produces $1.10^3 = 1.331$, or about a 33.1% rise. Using the midpoint measured current of 50.0 A, the corresponding values would be 57.9 A and 66.6 A respectively.

5. Expanded current model in terms of KV, voltage, diameter, pitch, and blade count

Substituting the loaded rotational speed relation $n = \lambda(K_V V/60)$ into the propeller current model gives a more informative expression.

$$I = (C_P \rho D^5 / \eta V) \cdot (\lambda K_V V / 60)^3$$

$$I = (C_P \rho D^5 \lambda^3 K_V^3 V^2) / (\eta 60^3)$$

$$I \propto C_P \lambda^3 K_V^3 V^2 D^5$$

This equation shows that, for comparable aerodynamic conditions, current scales with the cube of motor KV, the square of supply voltage, the fifth power of propeller diameter, and the cube of the loading factor. Pitch and blade count enter primarily through the power coefficient C_P.

$$I = (C_P \rho D^5 \lambda^3 K_V^3 (N_s V_c)^2) / (\eta 60^3)$$

$$I \propto K_V^3 N_s^2 D^5$$

6. How pitch and blade count enter the model

Pitch and blade count influence current through their effect on the propeller aerodynamic coefficients rather than through a simple universal exponent. A compact representation is given below.

$$C_P = f(J, P/D, \sigma, Re)$$

$$P/D = 4.0 / 9.0 = 0.444$$

$$\sigma = (N_b \bar{c}) / (\pi R)$$

If blade geometry is otherwise comparable, moving from a three-blade to a two-blade propeller reduces solidity in the ratio 2/3. This reduction tends to reduce blade profile drag and torque demand, but it also allows the motor to recover RPM, so the final current change is the result of competing effects rather than a single one-way rule.

7. Back-calculating the effective aerodynamic loading of the measured setup

For the present 9-inch propeller, the diameter in SI units is D = 0.2286 m. Assuming standard air density ρ = 1.225 kg/m^3 and defining the combined loading parameter $\Psi = C_P \lambda^3 / \eta$, the power law can be rearranged as follows:

$$\Psi = P_e / (\rho D^5 n_{nl}^3)$$

Using the measured power range and the no-load speed n_{nl} = 440 rev/s yields $\Psi_{min} \approx 0.01765$, $\Psi_{mid} \approx 0.01842$, and $\Psi_{max} \approx 0.01919$. This is useful because it compresses the unknown power coefficient, loading factor, and efficiency into a single experimentally anchored term.

8. Comparison with an earlier reference benchmark

A useful comparison benchmark is the earlier case of a 2812 900 KV motor at 22.4 V with a 10×5.0 three-blade propeller drawing 43.3 A. A same-family scaling relation for current can be written as follows:

$$(I_2 / I_1) \approx (K_{V2} / K_{V1})^3 (V_2 / V_1)^2 (D_2 / D_1)^5 (\lambda_2 / \lambda_1)^3 (C_{P2} / C_{P1}) (\eta_1 / \eta_2)$$

If efficiency and loading are first treated as similar and blade count is unchanged, then the dominant scaling gives (1100/900)^3 (24.0/22.4)^2 (9/10)^5 ≈ 1.238, implying approximately 53.6 A from the earlier 43.3 A benchmark. Adding a first-order pitch-ratio correction reduces the estimate to about 47.6 A, which aligns closely with the lower bound of the measured range. This comparison shows that the higher KV and voltage drive current upward, while the smaller diameter and lower pitch ratio strongly moderate that increase.

9. Parametric inflight current estimate

A simple documentation model for static-to-flight comparison is to define an unloading factor U_I such that $U_I = I_{air} / I_{static}$. If a notional inflight unloading band of 15% to 25% relative to static is assumed for comparative discussion, then U_I lies between 0.75 and 0.85.

$$I_{air} = U_I I_{static}$$

$$I_{air,min} = 0.75 \cdot 47.92 = 35.94 \text{ A}$$

$$I_{air,max} = 0.85 \cdot 52.08 = 44.27 \text{ A}$$

$$I_{air,15\%} = 0.85 \cdot 50.0 = 42.5 \text{ A}$$

$$I_{air,20\%} = 0.80 \cdot 50.0 = 40.0 \text{ A}$$

$$I_{air,25\%} = 0.75 \cdot 50.0 = 37.5 \text{ A}$$

At 24.0 V, these correspond to indicative inflight power levels of approximately 1020 W, 960 W, and 900 W respectively. These should be described as parametric inflight estimates derived from the static benchmark rather than as flight-test data.

10. Two-blade comparison

To compare a two-blade version of the same nominal 9×4.0 propeller, a first-order static current-ratio model can be written as:

$$I_{2b} / I_{3b} \approx (\lambda_{2b} / \lambda_{3b})^3 (\sigma_{2b} / \sigma_{3b})^b$$

$$\sigma_{2b} / \sigma_{3b} = 2 / 3$$

If RPM were held constant, the same-RPM current ratio would reduce to a pure solidity term. Using the three-blade midpoint current of 50.0 A gives the following same-RPM illustrations: b = 0.5 gives 40.8 A, b = 0.7 gives 38.9 A, and b = 1.0 gives 33.3 A. However, the two-blade propeller will generally allow the motor to spin faster. For b = 0.7 and a 5% RPM rise, the estimate becomes 43.6 A; for b = 0.7 and an 8% RPM rise, 47.4 A; and for b = 0.5 and a 10% RPM rise, 54.4 A. This shows that a two-blade propeller does not automatically guarantee lower current, because reduced solidity and RPM recovery act in opposition.

11. Summary table

Quantity	Equation / basis	Result	Interpretation
Measured static current	$I = P/V$	47.9–52.1 A	Directly measured electrical benchmark
Midpoint static current	(47.92 + 52.08)/2	50.0 A	Representative static point
No-load RPM upper bound	$RPM_{nl} = K_V V$	26400 rpm	Electrical upper limit before loading
RPM-current scaling	$I_2/I_1 = (n_2/n_1)^3$	cubic	Explains steep current rise with RPM
Combined loading parameter	$\Psi = P_e /(\rho D^5 n_{nl}^3)$	0.0177–0.0192	Measured setup compressed into one term
Simple scaled estimate vs earlier reference	$KV^3 \cdot V^2 \cdot D^5$ scaling	53.6 A	Slight overprediction without pitch correction
Pitch-corrected estimate	Scaled estimate × (0.444/0.500)	47.6 A	Very close to lower measured bound
Parametric inflight band	0.75–0.85 × static current	35.9–44.3 A	Illustrative unloading estimate
Two-blade same-RPM range	$(2/3)^b \times 50$ A	33.3–40.8 A	Potential reduction if RPM is unchanged
Two-blade with RPM recovery	$r^3 (2/3)^b \times 50$ A	43.6–54.4 A	RPM recovery can offset solidity reduction

12. Conclusion

The 2812 1100 KV, 24.0 V, 9×4.0 three-blade benchmark provides a measured static current of 47.9 A to 52.1 A, with a midpoint of 50.0 A at approximately 1200 W. Classical propeller power laws show that current scales approximately as $I \propto C_P \lambda^3 K_V^3 V^2 D^5$. This explains why higher KV and voltage raise current strongly, while even a modest reduction in diameter reduces current sharply because of the D^5 term. When

compared with an earlier 2812 900 KV, 22.4 V, 10×5.0 three-blade reference, the simple scaling model predicts about 53.6 A, whereas a first-order pitch-ratio correction reduces the prediction to about 47.6 A, which aligns closely with the measured current band. A parametric inflight unloading model suggests a documentation estimate of roughly 35.9 A to 44.3 A if the static benchmark is reduced by 15% to 25%. Finally, a two-blade comparison shows that lower blade count must always be analysed together with RPM recovery. The academically defensible conclusion is that current draw is governed by the coupled interaction of RPM, motor KV, supply voltage, propeller diameter, pitch ratio, blade solidity, and aerodynamic loading state.